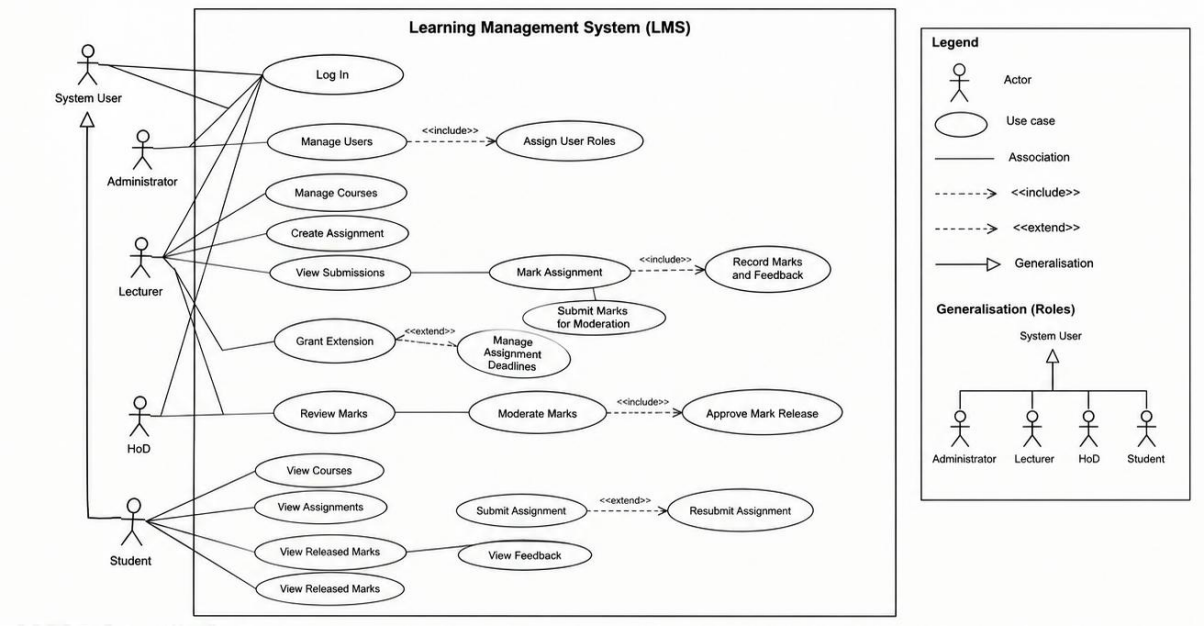
Use Case Model

Introduction

The use case model shows how the four main users interact with the Learning Management System (LMS). It shows the functions available to the Administrator, Lecturer, Head of Department (HoD) and Student. It also shows relationships between use cases, including <<include>>, <<extend>> and actor generalisation where appropriate.

Use Case Diagram



The diagram shows the main functions of the system and how each actor interacts with them. It also shows that marks must be moderated and approved before they are released to students.

Fully Dressed Use Case Specifications

Use Case 1: Submit Assignment

Use Case ID: UC-01
Primary Actor: Student
Description: Allows a student to submit an assignment through the LMS.

Preconditions

- The student is logged in.
- The assignment exists and is available to the student.
- The submission deadline has not passed, or an approved extension applies.

Main Flow

1. The student selects the assignment.
2. The system displays the assignment details and deadline.
3. The student uploads the assignment file and adds comments if needed.

4. The system checks the submission details and deadline.
5. The system saves the submission.
6. The system displays a confirmation message.

Alternative Flows

- If the submission is late but within the allowed three-day period, the system records it and applies the applicable 10% penalty per day.
- If the student has an approved extension, the system checks the new deadline instead of the original deadline.
- If the student submits another version before the applicable deadline, the system saves the new version and identifies it as the latest valid submission.

Exception Flows

- If the deadline has passed and no approved extension or late-submission period applies, the system rejects the submission.
- If the uploaded file cannot be processed or a system error occurs, the system displays an error message and does not confirm the submission as successful.

Postconditions

- The submission is saved and associated with the correct student and assignment.
- The student can view the submission status.

Business Rules

- Students may resubmit assignments before the applicable deadline.
- Late submissions receive a 10% penalty per day for up to three days, after which the mark is zero, unless an approved extension applies.

Use Case 2: Mark Assignment and Submit Marks for Moderation

Use Case ID: UC-02
Primary Actor: Lecturer
Description: Allows a lecturer to mark submitted assignments, provide feedback and submit the marks for moderation.

Preconditions

- The lecturer is logged in.
- The assignment exists.
- At least one submission is available for marking.

Main Flow

1. The lecturer selects the assignment.
2. The system displays the submitted work.

3. The lecturer reviews the submission.
4. The lecturer enters the mark and feedback.
5. The system validates and saves the mark and feedback.
6. The lecturer submits the marks for moderation.
7. The system records the submission for moderation and updates its status.

Alternative Flows

- If a submission is late, the system displays the applicable late-submission penalty.
- If the lecturer grants an individual extension, the system updates the deadline for the selected student and assignment.
- If the lecturer needs to correct a saved mark, the system records the change in the audit history.

Exception Flows

- If the mark is outside the permitted range, the system displays an error and requests a valid mark.
- If a system error occurs while saving, the system displays an error and does not confirm that the changes were saved.

Postconditions

- The mark and feedback are saved.
- The marks are submitted for moderation.
- The marks are not visible to students until they have been approved for release.

Business Rules

- Marks must be moderated by the HoD before they are released.
- All changes to marks must be recorded in the audit history.

Use Case 3: Moderate and Approve Marks for Release

Use Case ID: UC-03
Primary Actor: Head of Department (HoD)
Description: Allows the HoD to review marks submitted by lecturers and approve them for release to students.

Preconditions

- The HoD is logged in.
- Marks have been submitted for moderation by a lecturer.
- The relevant assessment records are available.

Main Flow

1. The HoD selects the assignment requiring moderation.
2. The system displays the marks and related feedback.
3. The HoD reviews the marks.
4. The HoD approves the marks for release.
5. The system records the moderation decision.
6. The system updates the marks' status to approved for release.
7. The system makes the approved marks and feedback available to the relevant students.

Alternative Flows

- If corrections are required, the HoD returns the marks to the lecturer with a reason for review and correction.
- After the lecturer makes the required corrections, the marks can be submitted for moderation again.

Exception Flows

- If the marks are incomplete, the system displays a message and prevents approval until the required information is available.
- If a system error occurs, the system displays an error and does not confirm the approval.

Postconditions

- If approved, the marks are recorded as approved for release and become visible to the relevant students.
- If returned for correction, the marks remain unavailable to students until they are approved.

Business Rules

- Marks must be moderated and approved before release.
- The system must record moderation decisions and mark changes in the audit history.
- Only an authorised HoD may moderate and approve marks for release.